

Homework 1

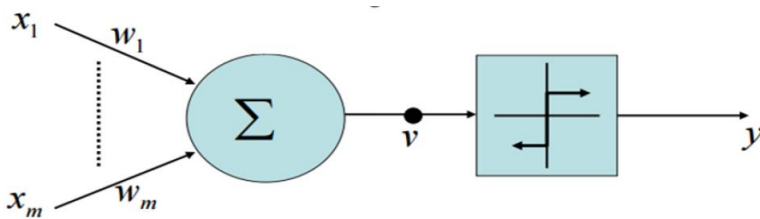
Problem A (20%)

A single-layer perceptron has an input vector $x = [0.5, -1.2, 0.3]$ and corresponding weights $w = [0.8, 0.4, -0.6]$ with no bias term. The perceptron uses a step activation function defined as:

$$f(v) = \begin{cases} 1, & \text{if } v \geq 0 \\ 0, & \text{if } v < 0 \end{cases}$$

Given that the desired output (target) $d = 1$, perform the following:

Figure 1.



(a) Compute the net input v and the perceptron output y . (10%)

✓ (b) If the output is incorrect, update the weights using the perceptron learning rule:

$$\Delta w_i = \eta \cdot (d - y) \cdot x_i$$

with learning rate $\eta = 0.1$, Provide the updated weights. (10%)

(a)

$$\begin{aligned} v &= 0.5 \times 0.8 + (-1.2) \times 0.4 + 0.3 \times (-0.6) \\ &= 0.4 - 0.48 - 0.18 \\ &= -0.26 \# \end{aligned}$$

$$y = f(v) = f(-0.26) = 0 \#$$

(b)

新的Weight:

$$\begin{cases} \Delta w_1 = 0.1 \cdot (1 - 0) \cdot 0.5 = 0.05 \\ \Delta w_2 = 0.1 \cdot (1 - 0) \cdot (-1.2) = -0.12 \\ \Delta w_3 = 0.1 \cdot (1 - 0) \cdot 0.3 = 0.03 \end{cases}$$

Problem B (80%)

Assume there is the multi-layer neural network in Fig. 1 is trained with a backward propagation algorithm and the current values of the weights are shown in the followed table. A vector $x = [0.3, 0.4, 0.1]_T$ serves as the input of the network and the result y is expected to be $[0, 1]_T$.

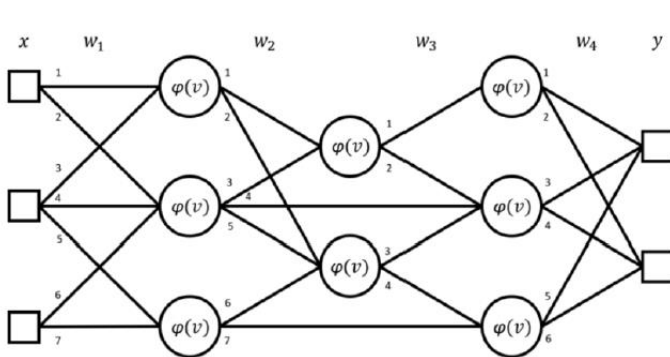
The learning rate $\eta = 0.3$.

The loss function $E = \frac{1}{2} (y_{pred} - y_{real})^2$

The activation function $\varphi(v) = \frac{1}{1 + \exp(-10v)}$. $\varphi'(v) = 10 \cdot \varphi(v) \cdot (1 - \varphi(v))$

Note that some values of the activation function are shown in Fig. 2.

Figure 1.



	w_1	w_2	w_3	w_4
1	0.2	0.5	0.1	0.6
2	0.2	-0.5	-0.9	0.3
3	0.2	0.5	0.9	-0.3
4	0.2	-0.5	-0.1	-0.9
5	0.2	0.5		0.1
6	0.2	-0.5		-0.1
7	0.2	0.5		

Figure 2.

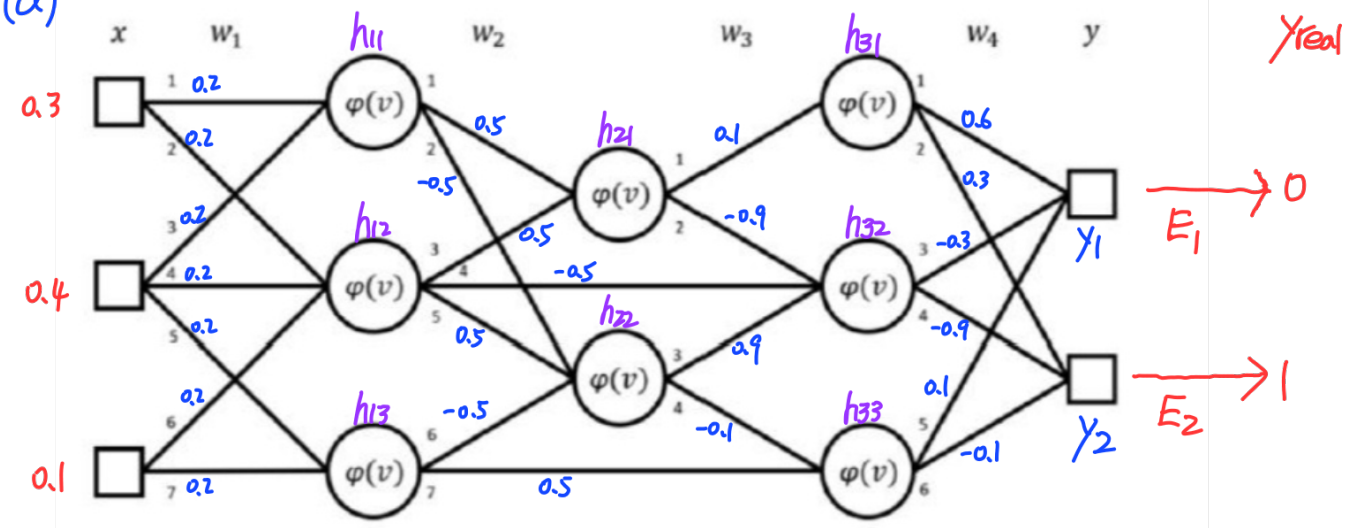
v	-0.53	-0.38	-0.35	0.1	0.14	0.16	0.35	0.38	0.40	0.53
$\varphi(v)$	0	0.02	0.02	0.73	0.8	0.83	0.98	0.98	0.98	1
$\varphi'(v)$	0	0.2	0.2	1.97	1.6	1.41	0.2	0.2	0.2	0

Questions:

- Please compute the value of each hidden node and y_{pred} . (20%)
- Please compute the value of the updated weights of w_3 after one iteration. (30%)
- Please compute the value of the updated weights of w_2 after one iteration. (30%)

(Deriving the equations first is recommended for the last two questions.)

(a)



令 h_{ij} 表示 hidden layer node

$h_{11} = \phi(0.3 \times 0.2) = \phi(0.06) = 0.8$, $h_{21} = \phi(0.4 \times 0.5) = \phi(0.2) = 1$, $h_{31} = \phi(0.1 \times 0.2) = 0.73$
 $h_{12} = \phi(0.3 \times 0.2) = \phi(0.06) = 0.83$, $h_{22} = \phi(0.4 \times 0.5) = \phi(0.2) = 0.02$, $h_{32} = \phi(0.1 \times 0.2) = 0.73$
 $h_{13} = \phi(0.3 \times 0.2) = \phi(0.06) = 0.73$, $h_{23} = \phi(0.4 \times 0.5) = \phi(0.2) = 0.98$

$$y_{pred} = \begin{bmatrix} 0.73 \times 0.6 + 0.98 \times 0.1 \\ 0.73 \times 0.3 + 0.98 \times (-0.1) \end{bmatrix} = \begin{bmatrix} 0.536 \\ 0.121 \end{bmatrix}$$

(b)

迭代次数
符号定义: $W_{ij}^t \rightarrow$ 权值位置 $V_{ij} \rightarrow \phi \rightarrow y_{ij}$

$$\frac{\partial E}{\partial y_{pred}} = (y_{pred} - y_{real}) = \begin{bmatrix} 0.536 \\ -0.879 \end{bmatrix}$$

$$\frac{\partial E}{\partial w_{31}} = \frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{31}} \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial w_{31}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{31}} \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial w_{31}}$$
$$= (\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{31}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{31}}) \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial w_{31}}$$
$$= (0.536 \cdot 0.6 + (-0.879) \cdot 0.3) \cdot \phi'(0.1) \cdot y_{21}$$
$$= (0.3216 - 0.2637) \cdot 1.97 \cdot 1$$
$$= 0.114063$$

$$\therefore W_{31}^1 = 0.1 - 0.3 \times 0.114063 = 0.0657811$$

$$\frac{\partial E}{\partial w_{33}} = \frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{33}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{33}}$$
$$= (\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}}) \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{33}}$$
$$= (0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) \cdot \phi'(-1.297) \cdot 0.02$$
$$= 0$$

$$\therefore W_{33}^1 = W_{33}^0 = 0.9$$

$$\frac{\partial E}{\partial w_{32}} = \frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{32}}$$
$$= (\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}}) \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial w_{32}}$$
$$= (0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) \cdot \phi'(-1.297) \cdot 1$$
$$= 0$$

$$\therefore W_{32}^1 = W_{32}^0 = -0.9$$

$$\frac{\partial E}{\partial w_{34}} = \frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial w_{34}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial w_{34}}$$
$$= (\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}}) \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial w_{34}}$$
$$= (0.536 \cdot 0.1 + (-0.879) \cdot (-0.1)) \cdot \phi'(0.363) \cdot 0.02$$
$$= 0.000566$$

$$\therefore W_{34}^1 = -0.1 - 0.3 \times 0.000566$$
$$= -0.1001698$$

$$\begin{aligned} \frac{\partial E}{\partial W_{21}} &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{31}} \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{31}} \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right) \right] + \\ &\quad \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right) \right] \\ &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{31}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{31}} \right) \frac{\partial y_{31}}{\partial v_{31}} \frac{\partial v_{31}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right] + \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \right) \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{21}} \frac{\partial y_{21}}{\partial v_{21}} \frac{\partial v_{21}}{\partial W_{21}} \right] \\ &= \left[(0.536 \cdot 0.6 + (-0.879) \cdot 0.3) \varphi'(0.1) \cdot 0.1 \cdot \varphi'(0.815) \cdot 0.8 \right] + \left[(0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) \cdot \varphi'(-1.297) \cdot (-0.9) \cdot \varphi'(0.815) \cdot 0.8 \right] \\ &= \left[0.0579 \cdot 1.97 \cdot 0.1 \cdot 0 \cdot 0.8 \right] + \left[0.6303 \cdot 0 \cdot (-0.9) \cdot 0 \cdot 0.8 \right] = 0 \end{aligned}$$

$$\begin{aligned}\frac{\partial E}{\partial W_{24}} &= \frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial W_{24}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial W_{24}} \\ &= \left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \right) \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial W_{24}} \\ &= (0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) \cdot (\varphi'(-1.297) \cdot 0.83) \\ &= 0 \\ \therefore W_{24}' &= W_{24}^0 = -0.5\end{aligned}$$

$$\begin{aligned} \frac{\partial E}{\partial W_{25}} &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial \beta_2} \frac{\partial \beta_2}{\partial V_{32}} \frac{\partial V_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial \beta_2} \frac{\partial \beta_2}{\partial V_{32}} \frac{\partial V_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right) \right] + \\ &\quad \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial \beta_3} \frac{\partial \beta_3}{\partial V_{33}} \frac{\partial V_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial \beta_3} \frac{\partial \beta_3}{\partial V_{33}} \frac{\partial V_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right) \right] \\ &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial \beta_2} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial \beta_2} \right) \frac{\partial \beta_2}{\partial V_{32}} \frac{\partial V_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right] + \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial \beta_3} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial \beta_3} \right) \frac{\partial \beta_3}{\partial V_{33}} \frac{\partial V_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial V_{22}} \frac{\partial V_{22}}{\partial W_{25}} \right] \\ &= \left[(0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) (\varphi'(-1.297) \cdot 0.9 \cdot \varphi'(-0.35) \cdot 0.83) \right] \\ &\quad + \left[(0.536 \cdot 0.1 + (-0.879) \cdot (-0.1)) (\varphi'(0.363) \cdot (-0.1) \cdot \varphi'(-0.35) \cdot 0.83) \right] \\ &= 0 + (1.415 \cdot 0.2 \cdot (-0.1) \cdot 0.2 \cdot 0.83) \\ &= -0.0046978 \\ &\therefore W_{25}^1 = 0.5 + 0.3 \times 0.0046978 = 0.50140934 \end{aligned}$$

$$\begin{aligned} \frac{\partial E}{\partial W_{26}} &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right) \right] + \\ &\quad \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right) \right] \\ &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{32}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{32}} \right) \frac{\partial y_{32}}{\partial v_{32}} \frac{\partial v_{32}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right] + \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}} \right) \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial y_{22}} \frac{\partial y_{22}}{\partial v_{22}} \frac{\partial v_{22}}{\partial W_{26}} \right] \\ &= \left[(0.536 \cdot (-0.3) + (-0.879) \cdot (-0.9)) \cdot \varphi'(-1.297) \cdot 0.9 \cdot \varphi'(-0.35) \cdot 0.73 \right] \\ &\quad + \left[(0.536 \cdot 0.1 + (-0.879) \cdot (-0.1)) \cdot \varphi'(0.363) \cdot (-0.1) \cdot \varphi'(-0.35) \cdot 0.73 \right] \\ &= 0 + [1.415 \cdot 0.2 \cdot (-0.1) \cdot 0.2 \cdot 0.73] \\ &= -0.0041318 \\ \therefore W'_{26} &= -0.5 + 0.3 \cdot 0.0041318 = -0.49876046 \end{aligned}$$

$$\begin{aligned}\frac{\partial E}{\partial W_{27}} &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial W_{27}} \right) + \left(\frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}} \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial W_{27}} \right) \right] \\ &= \left[\left(\frac{\partial E_1}{\partial y_1} \frac{\partial y_1}{\partial y_{33}} + \frac{\partial E_2}{\partial y_2} \frac{\partial y_2}{\partial y_{33}} \right) \frac{\partial y_{33}}{\partial v_{33}} \frac{\partial v_{33}}{\partial W_{27}} \right] \\ &= (0.536 \cdot 0.1 + (-0.879)(-0.1)) \cdot \varphi'(0.363) \cdot 0.73 \\ &= 0.1415 \cdot 0.2 \cdot 0.73 \\ &= 0.020659 \\ \therefore W_{27}' &= 0.5 - 0.3 \times 0.020659 = 0.4938023\end{aligned}$$